

80386 Operating Modes (Real, Protected & Virtual 8086) MPCA Module 5

> **Definition:** The 80386 microprocessor supports **three distinct operating modes**: **Real Address Mode** (16-bit 8086 compatible), **Protected Virtual Address Mode** (32-bit multitasking protected environment), and **Virtual 8086 (V86) Mode** (running real-mode 8086 software inside protected mode).

1. Detailed Comparison of 80386 Operating Modes

2. Deep Dive into Each Mode

1. Real Address Mode (Real Mode)

- Processor boots in this mode upon power-up/reset.

2. Protected Virtual Address Mode (Protected Mode)

- Activated by setting **PE = 1** in Register CR0.

3. Virtual 8086 Mode (V86 Mode)

- Sub-mode of Protected Mode enabled by setting **VM = 1** in EFLAGS register.

3. Operating Mode Comparison Table

| Property | Real Mode | Protected Mode | Virtual 8086 Mode |

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4. Quick Recall Flow

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Power
Mode